MILITARY STANDARD

CARLOADING

BOMB GUIDANCE KIT
KMU-351 A/B (LESS SEEKER ASSEMBLY)
FLEET ISSUE UNIT LOAD

MIL-STD-1325-163
(NAVY)

27 MARCH 1973

UNIT LOAD DATA

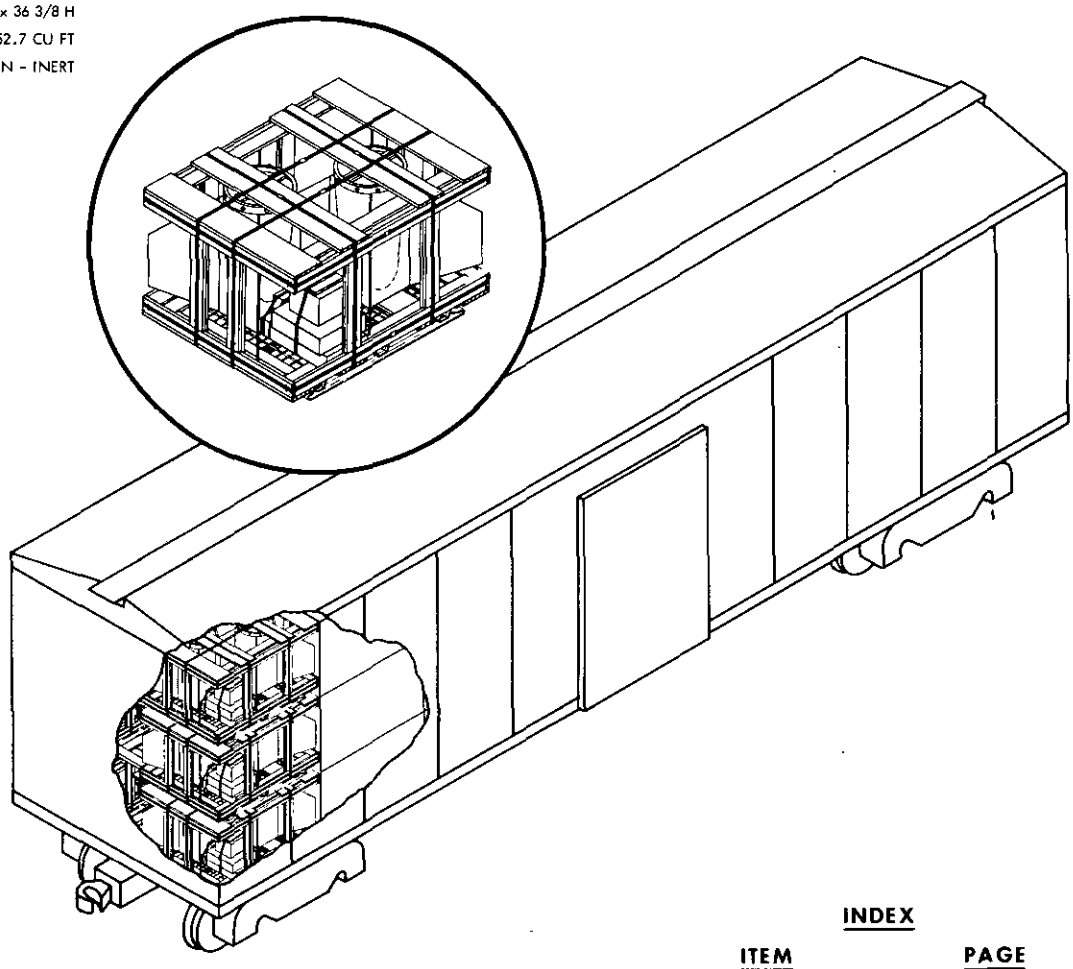
UNIT LOAD DRAWING . . . WR-54/236

WEIGHT . . . . . 484 LBS

DIMENSIONS . . 57 L x 44 W x 36 3/8 H

CUBE . . . . . 52.7 CU FT

DOT HAZARD CLASSIFICATION - INERT



1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINER LOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

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40' 6" CARLOAD, COML	6, 7 & 8
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AUTHORIZED AND RELEASED FOR GENERAL USE.

M. S. Gray NWHL 3/26/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 3/26/73
SIGNATURE (AIR) SYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

NOT APPLICABLE

MILITARY ASSISTANT

SIGNATURE DATE

ORIGINATOR Charles McBride 3/26/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF BOMB GUIDANCE KITS, KMU-351A/B (LESS SEEKER ASSEMBLY) PALLETIZED IN ACCORDANCE WITH WR-54/236.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. APPLICABLE MATERIAL SPECIFICATIONS:

   DUNNAGE LUMBER - FED. SPEC MM-L-751

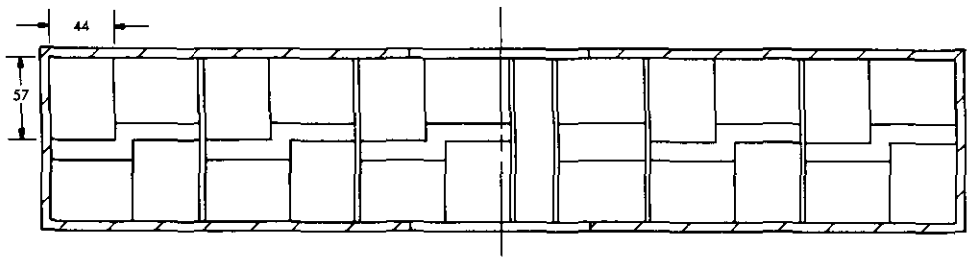
   NAILS - FED. SPEC FF-N-105

   STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

   SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
10. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL I).

50 FT 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 66 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSS MEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
8	LATERAL FLOOR MEMBER	2 x 6 x 107	2	CAR FLOOR	2 PER FOOT	16d
7	CROSS BRACE	2 x 6 x W*	6	SEE 6	-	-
6	SWAY BRACE LONGITUDINAL	1 x 4 x 56	4	5, 7	SEE NOTE **	
5	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	4	SEE 6	-	-
4	SLEEPER	2 x 4 x 54	2	CAR FLOOR	2 PER FOOT	16d
3	SWAY BRACE HORIZONTAL	2 x 4 x 57	60	SEE 1	-	-
2	SWAY BRACE VERTICAL	1 x 4 x 110	30	3	2 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 x 4 x 110	30	3	2 PER JOINT	16d

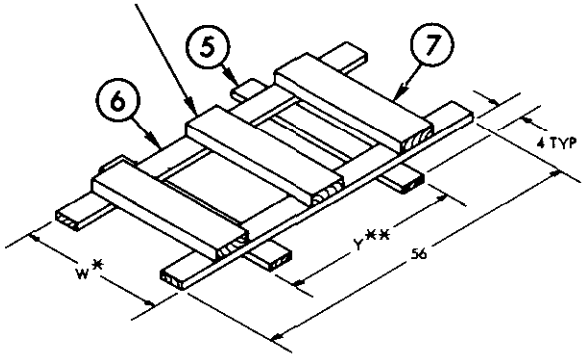
LIST OF MATERIALS AND NAILING DATA

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** PIECE 6 NAILS TO PIECE 7 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 5 WITH 6d NAILS, 1 PER JOINT

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CROSS BRACES MUST BE CENTERED AGAINST LEGS OF PALLETS

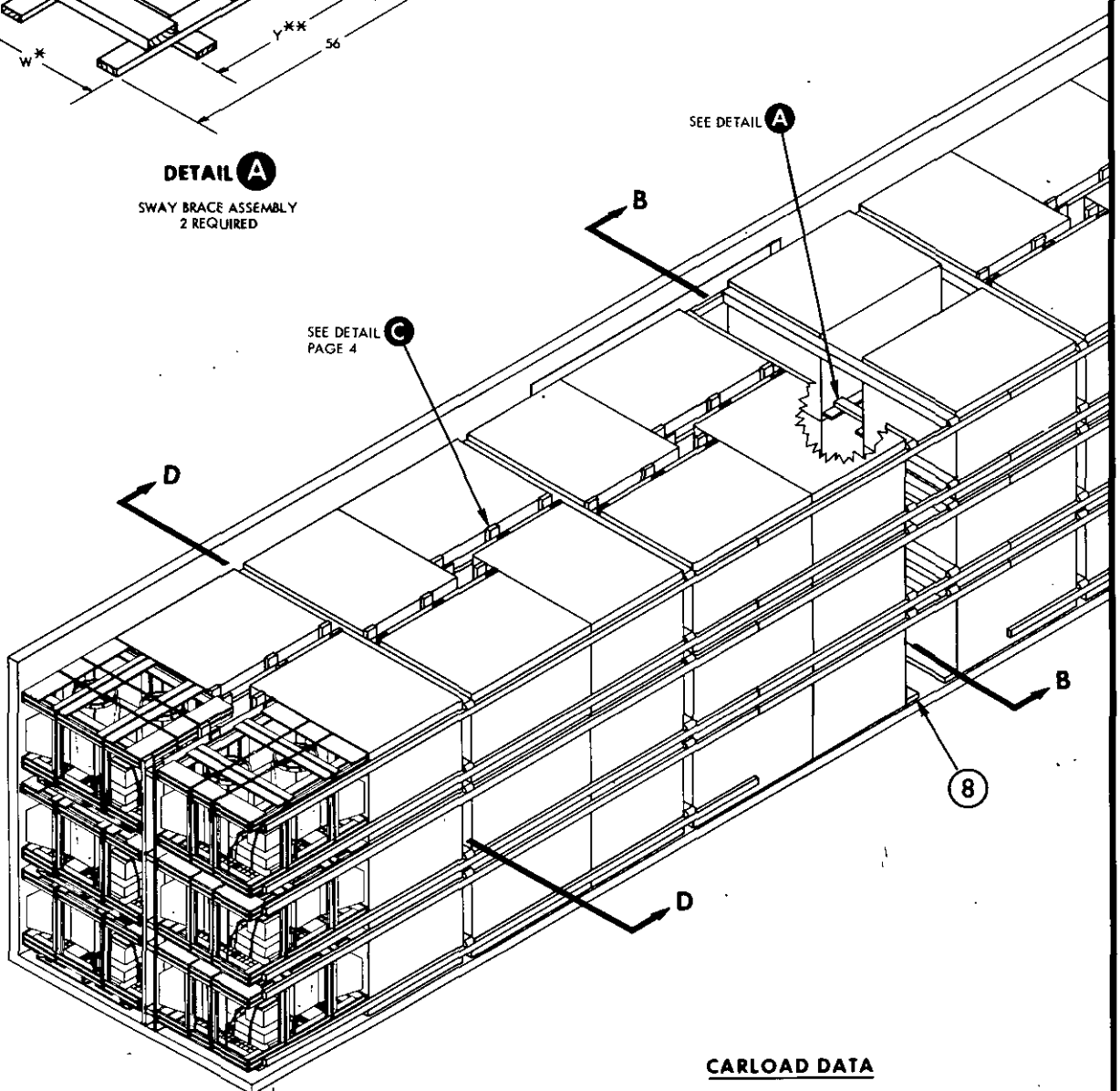


*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

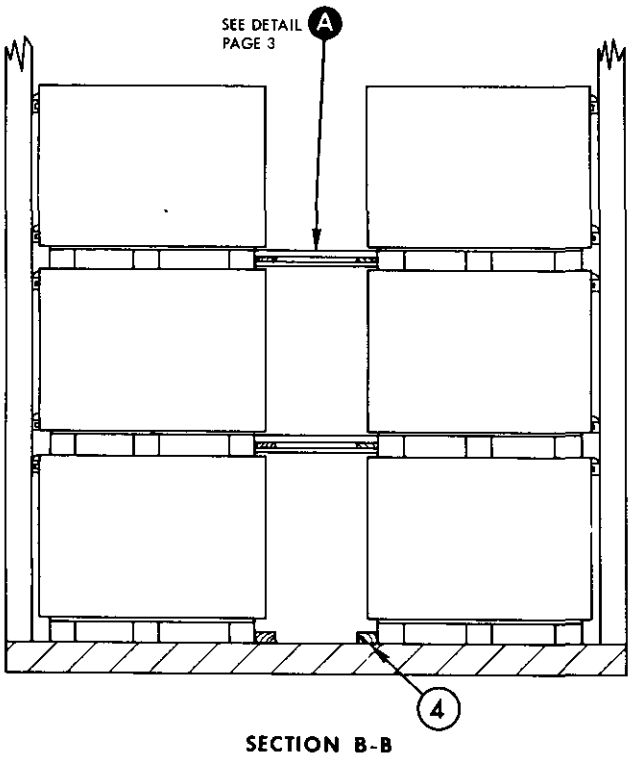
SWAY BRACE ASSEMBLY
2 REQUIRED



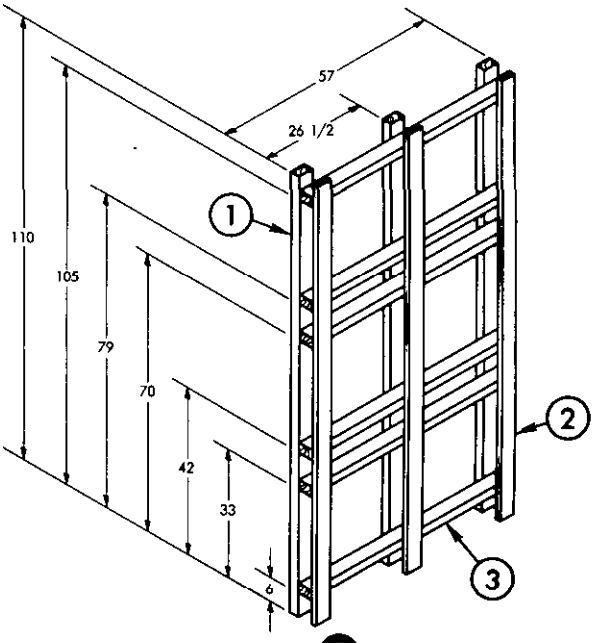
CARLOAD DATA

NUMBER OF UNIT LOADS	66
NUMBER OF CROSS MEMBERS REQ'D	34
LOAD WEIGHT	31,944 LBS
DUNNAGE WEIGHT	1,273 LBS
CARLOAD WEIGHT	33,217 LBS

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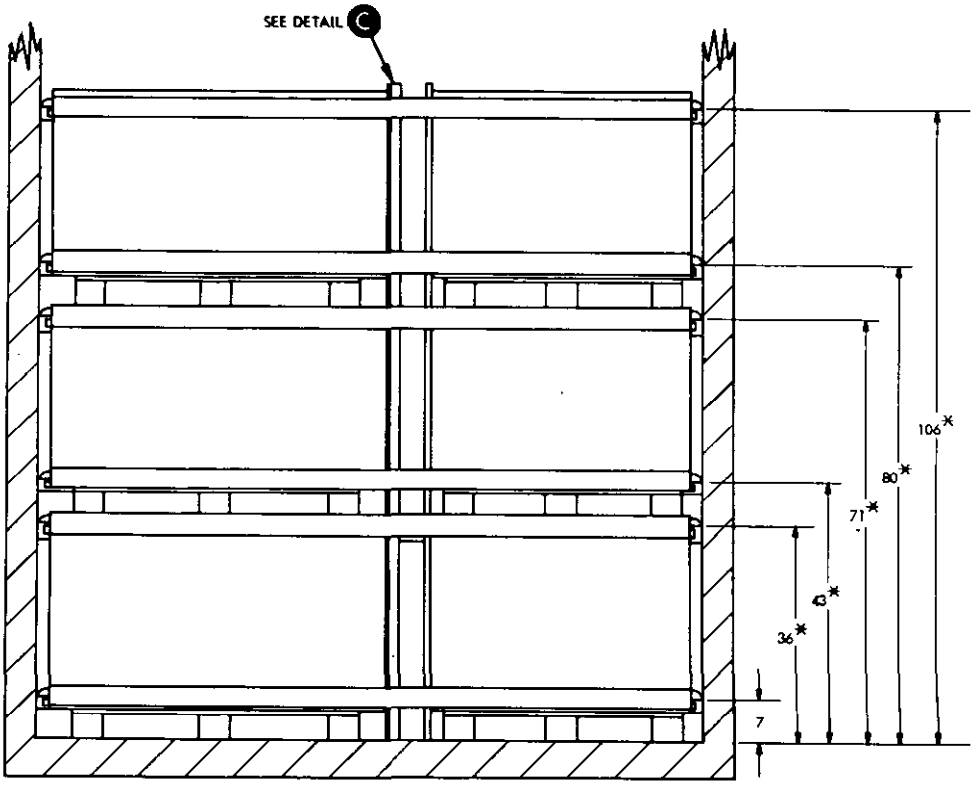


SECTION B-B



DETAIL C

SWAY BRACE ASSEMBLY
10 REQUIRED



SECTION D-D

SHOWING LOCATION OF WALL MEMBERS
USED FOR CROSS MEMBERS

* DOORWAY MEMBER LOCATIONS

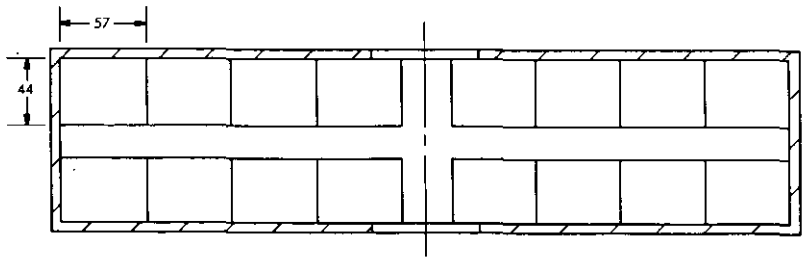
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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 48 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
11	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	10	2 PER JOINT	16d
10	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
9	STRUT	4 x 4 x WEDGE FIT ***	16	6	2 PER JOINT	16d
8	GATE HOLD DOWN	2 x 4 x 54	2	7	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 112	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	48	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 54	32	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	32	SEE 3	-	-
1	SLEEPER	2 x 4 x 54	16	CAR FLOOR	2 PER FOOT	16d

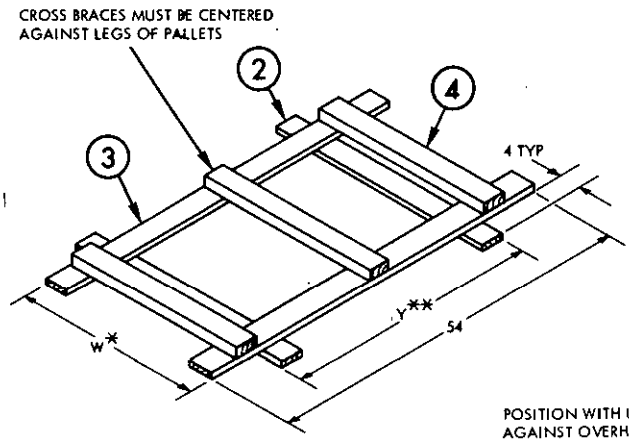
LIST OF MATERIALS AND NAILING DATA

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

*** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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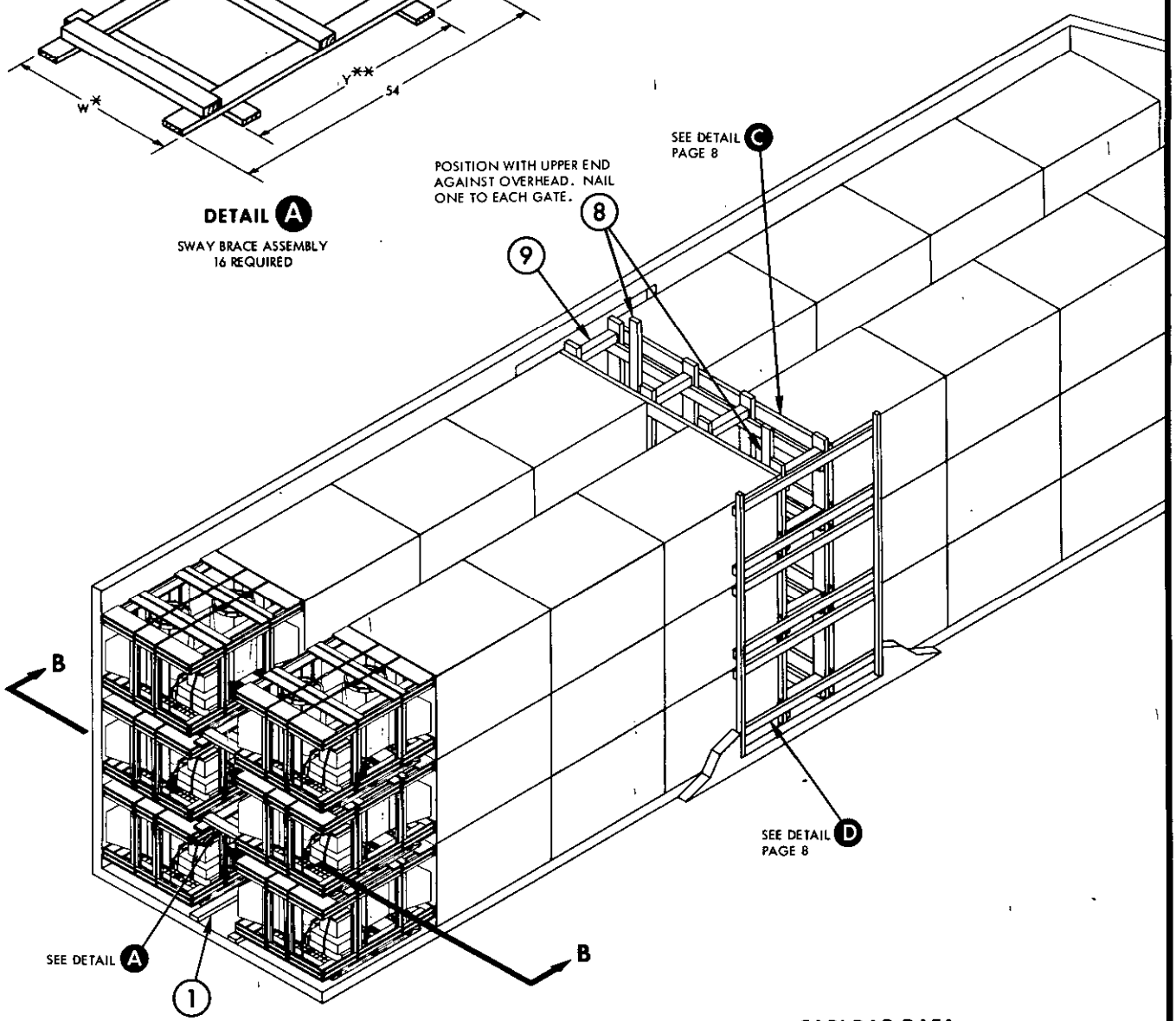
* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

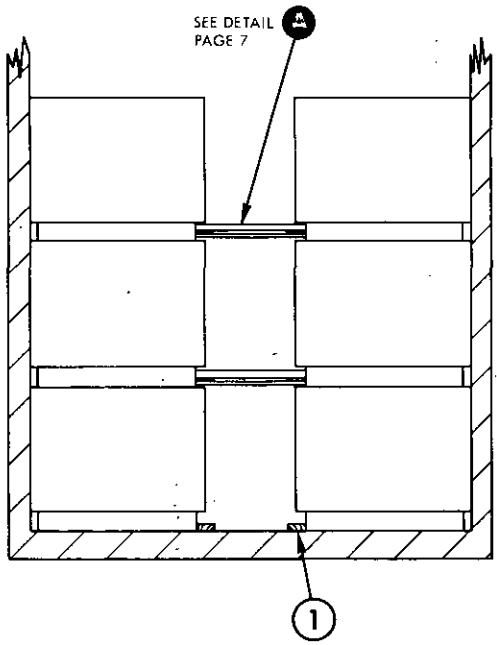
SWAY BRACE ASSEMBLY
16 REQUIRED



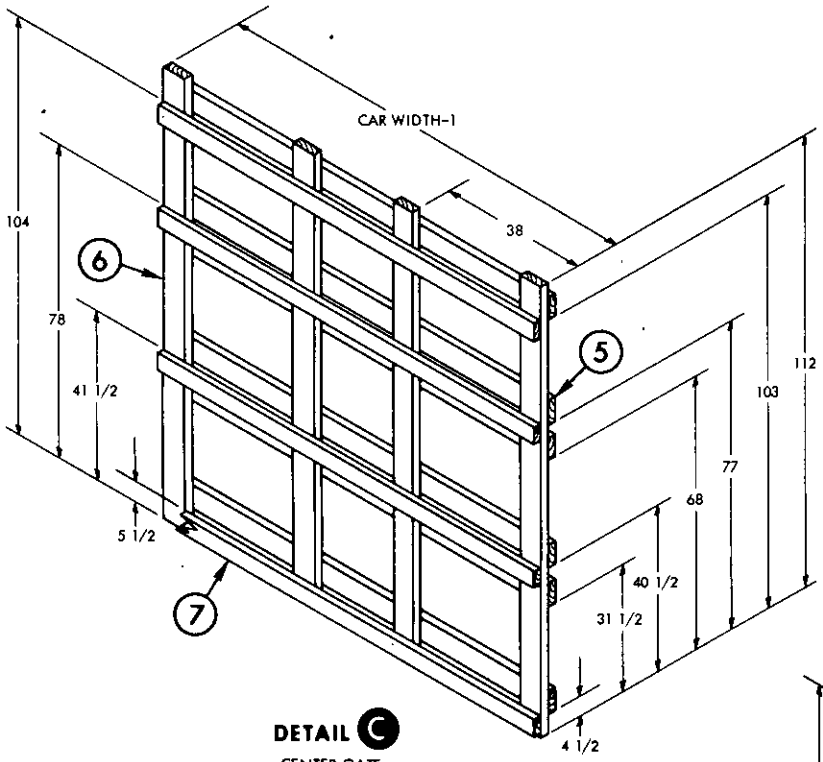
CARLOAD DATA

NUMBER OF UNIT LOADS	48
LOAD WEIGHT	23,232 LBS
DUNNAGE WEIGHT	1,426 LBS
CARLOAD WEIGHT	24,658 LBS

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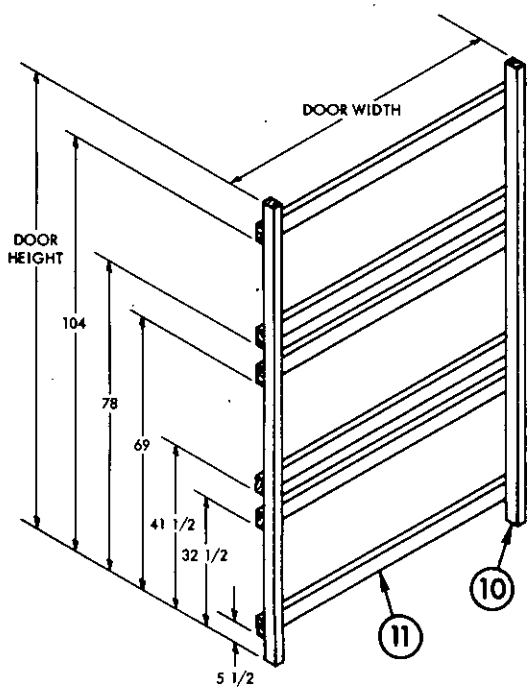


SECTION B-B



DETAIL C
CENTER GATE
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS
DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY)
PAGE 46, FIGURE 21 IS REQUIRED



DETAIL D
DOORWAY PROTECTION
2 REQUIRED

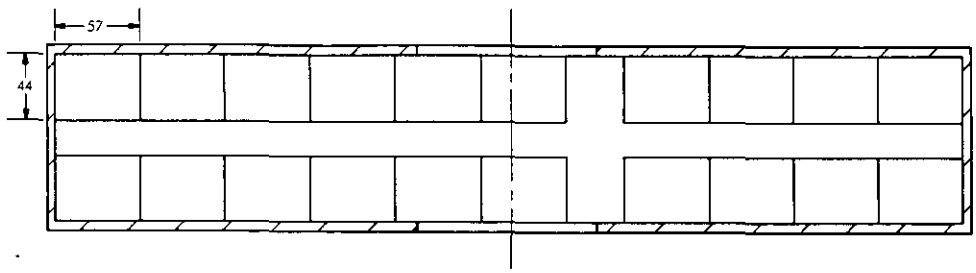
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. *THE CARLOAD CONSISTS OF 60 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL* DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D), MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 10 AND 11 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
11	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	12	10	2 PER JOINT	10d
10	VERTICAL DOORWAY MEMBER	2 x 3 x DOOR HEIGHT	4	DOOR POST	2 PER FOOT	20d
9	STRUT	4 x 4 x WEDGE FIT ***	16	6	2 PER JOINT	16d
8	GATE HOLD DOWN	2 x 4 x 54	2	7	3 PER JOINT	10d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 112	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	12	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	60	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 54	40	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	40	SEE 3	-	-
1	SLEEPER	2 x 4 x 54	20	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

* W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

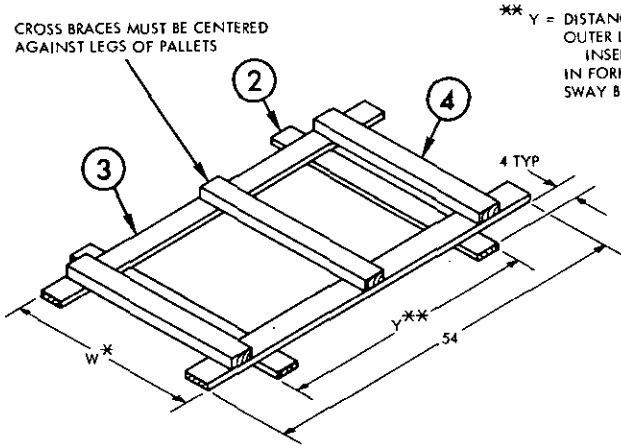
** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT

*** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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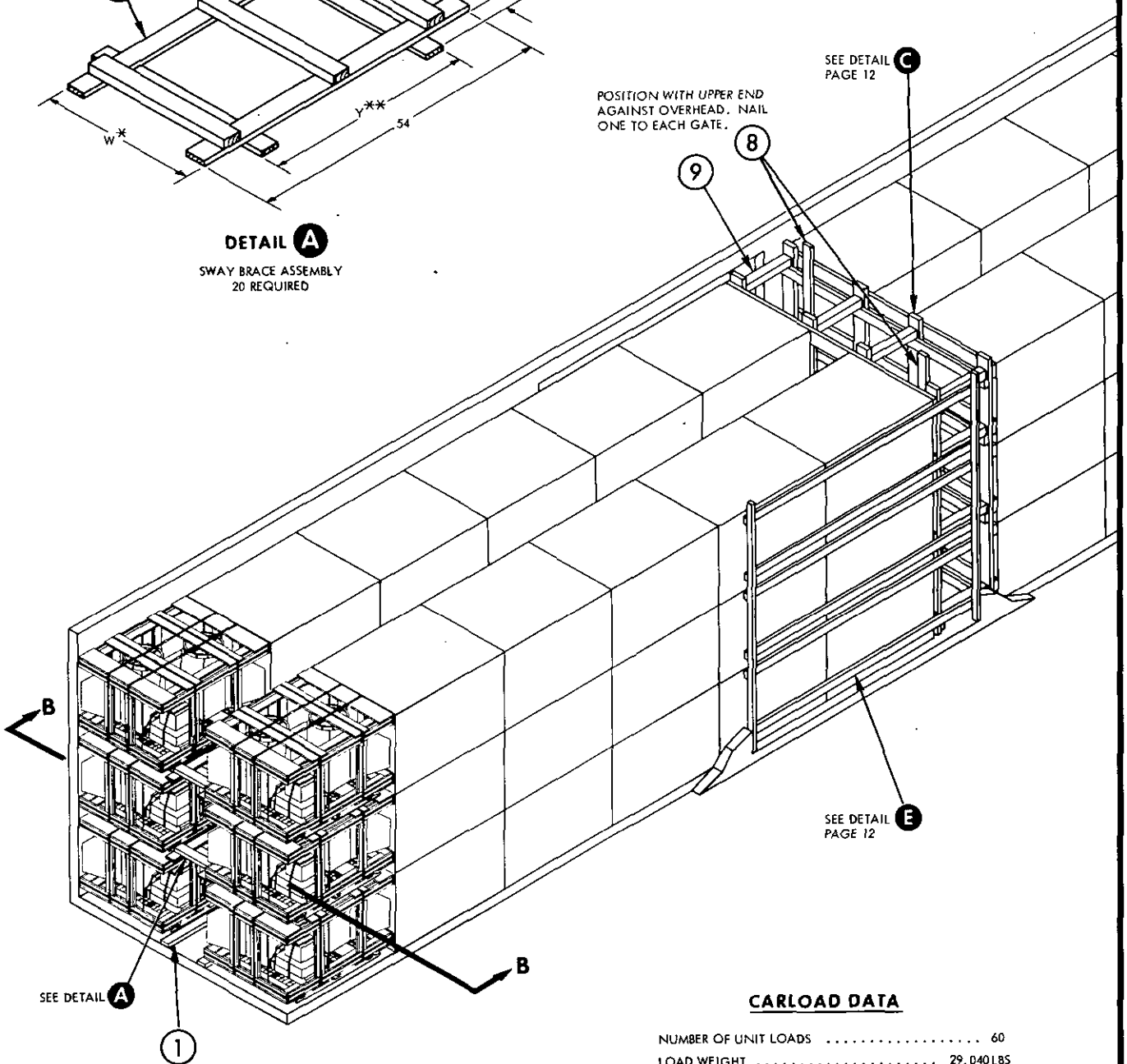
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET. INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

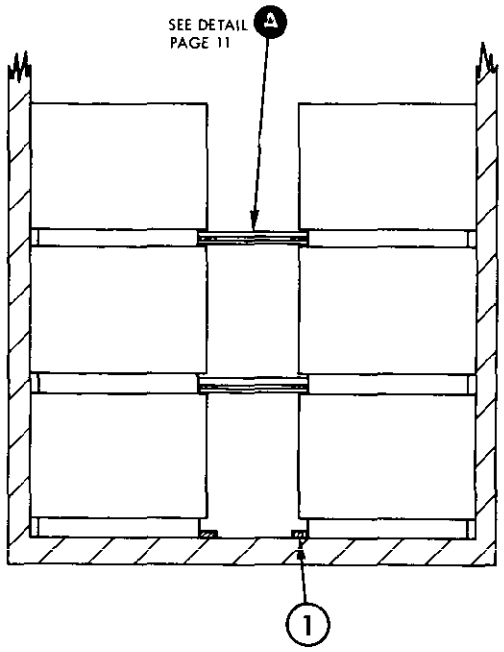
SWAY BRACE ASSEMBLY
20 REQUIRED



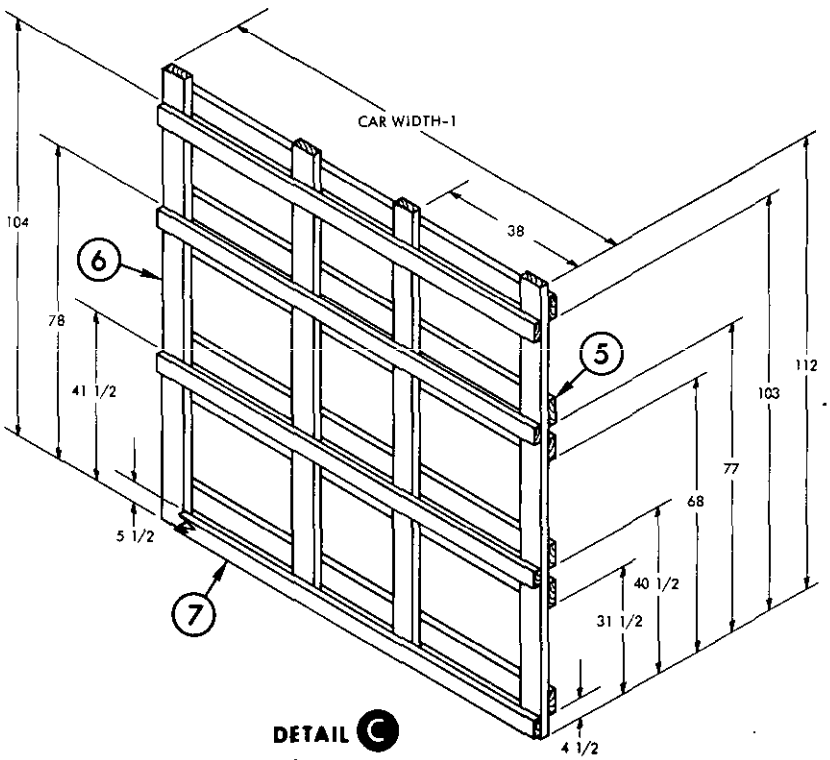
CARLOAD DATA

NUMBER OF UNIT LOADS	60
LOAD WEIGHT	29,040 LBS
DUNNAGE WEIGHT	1,663 LBS
CARLOAD WEIGHT	30,703 LBS

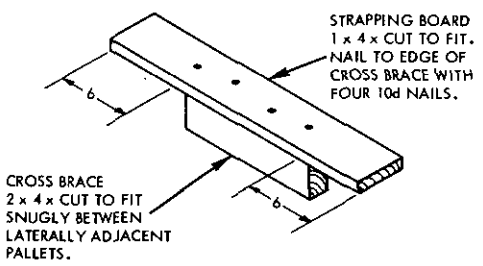
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SECTION B-B

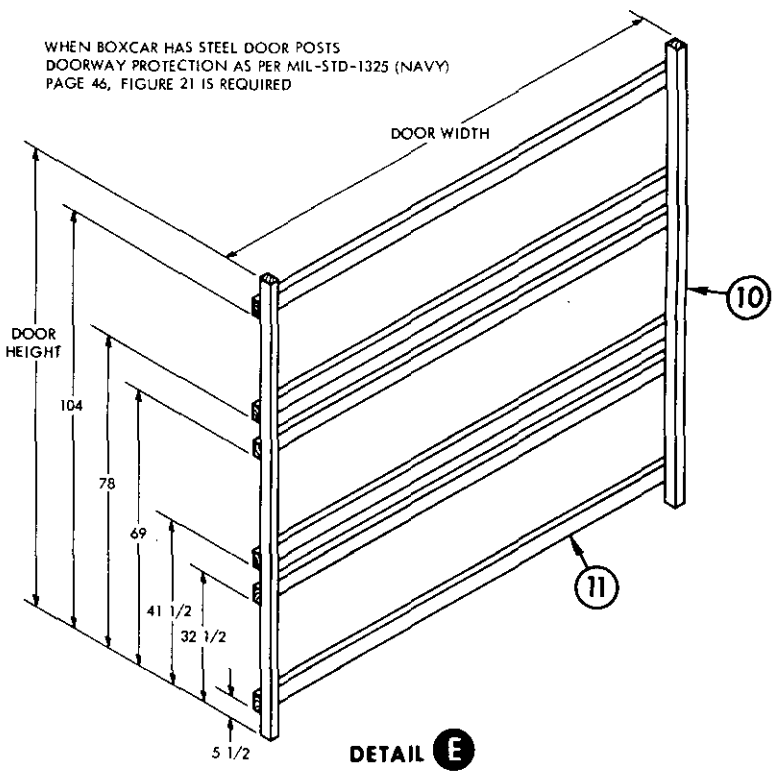


DETAIL C
CENTER GATE
2 REQUIRED



DETAIL D
TOP SWAY BRACE

WHEN BOXCAR HAS STEEL DOOR POSTS
DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY)
PAGE 46, FIGURE 21 IS REQUIRED



DETAIL E
DOORWAY PROTECTION
2 REQUIRED

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